



# Accurate prediction of discontinuous crack paths in random porous media via a generative deep learning model

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Edited by Yonggang Huang, Northwestern University–Evanston, Glencoe, IL; received July 7, 2024; accepted August 29, 2024

Pore structures provide extra freedoms for the design of porous media, leading to desirable properties, such as high catalytic rate, energy storage efficiency, and specific strength. This unfortunately makes the porous media susceptible to failure. Deep understanding of the failure mechanism in microstructures is a key to customizing high-performance crack-resistant porous media. **However, solving the fracture problem of the porous materials is computationally intractable due to the highly complicated configurations of microstructures.** To bridge the structural configurations and fracture responses of random porous media, a unique generative deep learning model is developed. A two-step strategy is proposed to deconstruct the fracture process, which sequentially corresponds to elastic deformation and crack propagation. The geometry of microstructure is translated into a scalar of elastic field as an intermediate variable, and then, the crack path is predicted. The neural network precisely characterizes the strong interactions among pore structures, the multiscale behaviors of fracture, and the discontinuous essence of crack propagation. Crack paths in random porous media are accurately predicted by simply inputting the images of targets, without inputting any additional input physical information. The prediction model enjoys an outstanding performance with a prediction accuracy of 90.25% and possesses a robust generalization capability. The accuracy of the present model is a record so far, and the prediction is accomplished within a second. This study opens an avenue to high-throughput evaluation of the fracture behaviors of heterogeneous materials with complex geometries.

random porous media | discontinuous crack paths | strong interactions | deep learning model

Porous media, such as ceramics, porous carbon, and metal foams (1–3), have attracted intensive attention and are widely used in catalytic engineering (4, 5), electrodes for energy storage (2, 6), and lightweight design of aircraft (7, 8). To obtain excellent properties, the microstructures of porous media are usually designed into complex configurations (9, 10). However, the porous structures make porous media susceptible to failure (11, 12). Fracture of microstructures is a key factor dominating the service life of these materials. Predicting the crack propagation in microstructures is consequently crucial to material designs (13). Unfortunately, capturing the fracture behavior in complicated geometries is an extremely difficult task owing to the strong interactions among pores (*SI Appendix, Fig. S1*) (14–16). **Physical-based numerical approaches to fracture solution, such as finite element method and discrete element method (17–19), are usually limited by finite computational resources.** Numerical modeling of the fracture problems requires considerations of the complex nonlinear mechanical responses on large spatial and temporal scales (20). A single simulation usually takes hours to days, depending upon the resolution of the numerical model (17). The computational cost is unacceptably expensive, especially for porous media with random geometries.

Deep learning has been proven a powerful tool to excavate inherent rules and complex correlations from a large amount of data (21). In view of the overwhelming advantages in skipping tedious physical computations (22–24), the deep learning method is a promising approach to address fracture problems of complex microstructures (25, 26). The propagation of continuous cracks (27) in solids can be predicted by time-series-dependent models, such as Convolutional Long-Short Term Memory models (28). This methodology infers the direction of the crack evolution by analyzing the continuous spatiotemporal information (29, 30) and is consequently inadequate for jump-type problems, such as discontinuous crack propagation in porous media in question in the present study. By means of the computer vision task in the context of the classical convolutional neural network (CNN) architecture (31), the ultimate crack paths can be predicted directly from the input images without tracking the process of crack propagation. However, porous media usually possess highly complicated geometric configurations (Fig. 1*A*). Owing to the strong interaction

## Significance

**Porous media are susceptible to fracture.** A deep understanding of crack propagation in porous microstructures remains limited to date, owing to the **unacceptably expensive cost of numerical modeling for fracture problem of highly complicated geometries.** To bridge this gap, we developed a unique deep learning model to predict the crack paths in random porous media directly from their images. Tedious physical simulations and experimental processes are skipped. The prediction time of the fracture is shortened from hours to a second in comparison with the traditional numerical models. The model enjoys high accuracy and robust generalization capability. This methodology provides perspectives of high-throughput evaluation of the fracture behaviors of the complex heterogeneous materials, consequently accelerating the design of high-performance crack-resistant materials.

Author contributions: X.L. designed research; Y.H., Y.T., M.Y., Y.W., and X.L. performed research; Y.H., Y.T., X.L., W.C., and G.K. analyzed data; and Y.H., Y.T., Y.X., J.Y., X.L., W.C., and G.K. wrote the paper.

The authors declare no competing interest.

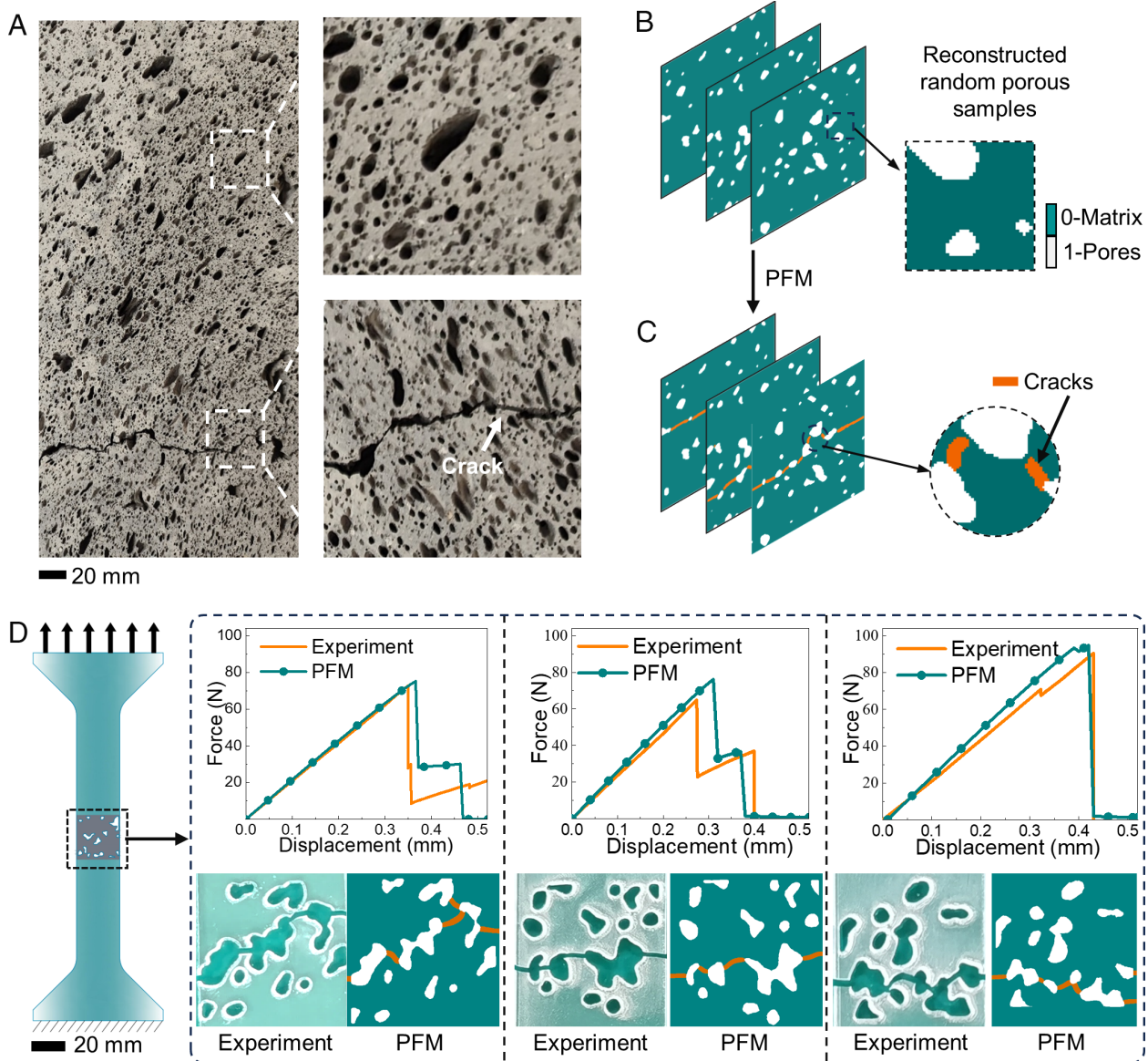
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This article contains supporting information online at <https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2413462121/-DCSupplemental>.

Published September 25, 2024.



**Fig. 1.** Fracture of random porous media. (A) Profiles of porous basalt. The pores with irregular geometries are randomly distributed in a matrix. (B) Reconstructed random porous samples based on the porous structures of basalt. The “white” and “dark blue” colors represent the pores and matrix, respectively. (C) Crack paths of random porous media samples obtained from the PFM. (D) Fracture experiments with porous polymethacrylate (PMMA) sheets and the corresponding fracture behaviors.

among the microstructures, the fracture of porous media is sensitive to local geometric features (32–34). For example, the crack path of a porous sheet can be completely altered even by changing a few elements in the local pore (*SI Appendix, Fig. S2*). The challenge of predicting crack propagation in porous media is evidenced. The reported works on deep learning modeling for prediction of crack path to date are only suitable for simple fracture patterns (35, 36) or rely on some prior conditions, such as a definite initial crack propagation point (37) or a given physical field (38).

In this study, a generative deep learning model is developed to predict crack propagation in porous sheets from the images of microstructures. The complex fracture process is split into two substeps. First, the image of microstructure is translated into the elastic strain energy density (SED) field. It works as an intermediate variable, bridging the geometry of microstructure and fracture response. The neural network then infers the crack path based on the predicted image of SED. The present model precisely characterizes the strong interactions among the micropores by extracting their multiscale feature maps. It works independently

without any assistance of the prior physical information. The discontinuous crack paths in complex porous media can be predicted within a second. The accuracy reaches more than 90%, which is a record so far for random porous media. The present model avoids expensive computations of fracture mechanics and significantly improves the evaluation efficiency of fracture responses in complicated porous media.

## Results

**Fracture of Porous Sheets.** We focus on the fracture of brittle porous sheets with randomly distributed micropores. Porous basalt with naturally generated pores is a representative example of this class of materials (Fig. 1A). To numerically reconstruct such porous samples, a so-called level-cut Gaussian random field method (39) is adopted. The reconstructed samples involved in each scenario have different microgeometries but possess the same statistical characteristics, as shown in Fig. 1B and *SI Appendix, Fig. S3*.

The reconstructed samples are fractured under the vertical tensile displacement loading. Owing to the highly stochastic nature of the geometric configurations, the strong interactions among the microstructures dominate the mechanical behaviors of the porous sheet. The matrix around each pore of the porous sheet is under a complex stress state in response to a simple uniaxial tensile displacement loading. Hence, the cracks propagate with the mixed mode (I and II). Furthermore, the crack paths propagate in the discontinuous way in porous sheets. A series of discontinuous short cracks are generated in the matrix and then form a complete crack path (Fig. 1C). The phase-field model (PFM) (40) is employed to simulate this fracture process. To validate the PFM, polymethacrylate (PMMA) sheets with random pores are fabricated. Their fracture responses are simultaneously simulated by the PFM and tested by experiments. As shown in Fig. 1D, the crack paths and force–displacement curves of the three reconstructed porous sheets can be accurately reproduced by the PFM. Therefore, the images of the elastic field and crack paths obtained by the PFM can serve as the ground truth in the deep learning model.

**Development of Deep Learning Model.** A two-step strategy is adopted in this work. The fracture process with complex nonlinear mechanical behavior is decomposed into two substeps, namely the stages of elastic deformation and crack propagation. The SED of elastic deformation acts as an intermediate scalar variable to connect the microstructure and final crack path. Accordingly, the developed deep learning framework consists of two generative networks, i.e., Generator 1 and Generator 2, as shown in Fig. 2A. For Generator 1, the images of initial geometries and the elastic SED field serve as input and output, respectively. Generator 2 then translates the predicted SED field from Generator 1 into the images of crack paths. To train the generative networks, discriminative network is simultaneously trained for each generative model. Each pair of Generator and Discriminator constitutes a conditional Generative Adversarial Net [cGAN (31)], as shown in Fig. 2B.

Both Generator 1 and Generator 2 possess the same U-Net 3+ architecture (41). This architecture is an encode–decode CNN with internal full-scale skip connections as shown in Fig. 2C. In view of the multiscale nature of fracture, the shallow convolutions of the encoding layers are used to capture the local features and the deeper networks concentrate on the global information pertinent to the porous structures. The full-scale skip connections provide the feature maps at all scales for each layer of the decoder. Therefore, the Generators can directly extract and fuse the fine-grained and coarse-grained semantics from the input images. In this manner, the present neural network exactly characterizes the strong interactions among the microstructures and learns the multiscale fracture mechanism of porous media.

Similarly, the Discriminator 1 and Discriminator 2 employ the same PatchGAN framework (31) (Fig. 2D). During the training process, the Generator and Discriminator are trained simultaneously and play against each other. The Generator aims to generate fake pictures as much as possible to deceive the Discriminator, while the objective of Discriminator tries to distinguish the image generated by Generator from its real counterpart (PFM results). The training is terminated when the game of Generators and Discriminators reaches an equilibrium. Once the prediction model is available, crack path of a porous sheet may be predicted within a second directly from its image.

**Model Performance.** The representative predictions for two test samples are shown in Fig. 3A. The predicted SED field and crack paths are in perfect agreement with the PFM results from an overall point of view. The stress concentrations in the vicinity

of the pores are accurately reproduced. The comparison of the predicted SED and the corresponding PFM counterparts is detailed in SI Appendix, Fig. S4. To evaluate the accuracy of predicted SED field, the mean absolute errors (MAEs) for 400 test samples are calculated and shown in Fig. 3B. The MAEs for most of the test samples are less than 0.030, and their average is 0.015. The excellent performance of Generator 1 lays a solid foundation of ensuingly predicting the crack paths.

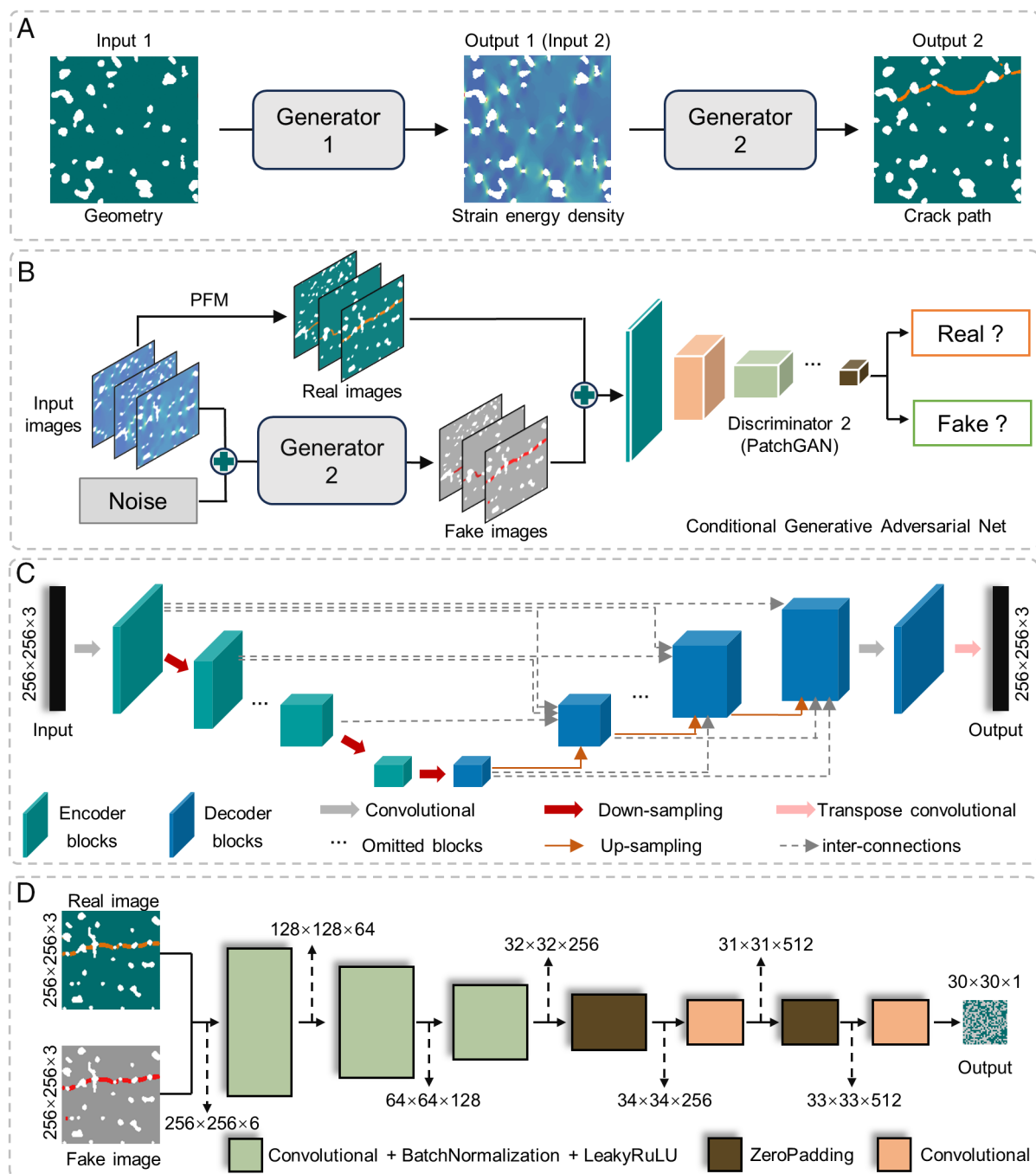
To quantify the similarity of crack paths between the ground truth and prediction result, a crack detection algorithm in terms of the modified Hausdorff distance is introduced (42). The fundamental idea is graphically shown in SI Appendix, Fig. S5. This method is detailed in SI Appendix, Text. With this crack detection algorithm, a score between 0 and 100 is assigned to each test sample. A higher score indicates a closer distance between the predicted and real crack paths. In this study, the predictions are separated into three categories namely Excellent (score of 70 ~ 100), Acceptable (score of 50 ~ 70), and Incorrect (score of 0 ~ 50). Both Acceptable and Excellent categories are defined as correct predictions. Some representative predictions of crack paths and the corresponding scores are illustrated in Fig. 3C. It is seen that the Incorrect predicted crack paths are quite different from the real content. When the score increases to 57.99, the predicted crack path exhibits a good consistency with the real one from an overall point of view. When the score is greater than 70.00, the predicted crack path is approximately identical to the true behavior, and most of the segments of the predicted crack coincide with the real crack counterpart. More detailed comparisons of crack paths of those samples are detailed in SI Appendix, Fig. S6.

The statistical results of the score for all the 400 test samples are presented in Fig. 3D. The mean score reaches 81.58, and a high accuracy of 90.25% is achieved, including Excellent and Acceptable predictions. To be more precise, 77.75% of the test samples belong to Excellent predictions, 12.50% of predictions correspond to Acceptable results, and the remaining 9.75% predictions fall into the Incorrect category. The present model enjoys an outstanding performance. It should be highlighted that the foregoing predictions are performed without any physical information as a prior. The constraint on the crack initiation point in the literature (37) is removed. The auxiliary variable SED is obtained by the Generator 1 instead of an inputting parameter as required in the literature (38).

**Generalization Capability.** To further test the capability of the present deep learning model, it is applied to some test samples with unique porous patterns. The customized test samples include patterns of “Snowflakes” and “Trees.” The shapes of their pores are regular rather than stochastic. The predicted results and corresponding numerical results are presented in Fig. 4A. Excellent agreements are again achieved. Furthermore, 400 porous samples with randomly distributed circular pores are generated and tested. The present model also works well with mean scores 78.67 and overall accuracy 89.00%, respectively. The representative predictions and the corresponding statistical histogram of the scores of predicted crack path are shown in Fig. 4B and C, respectively. The geometric features of these test samples under consideration have never been involved in the training data. It indicates a robust generalization capability of the present model.

**Application to Experimental Images.** The performance of the proposed model in predicting crack paths in complex porous structures has been systematically validated in the previous subsections. In this subsection, two cases are considered to demonstrate the engineering applications of the present model. The geometries of the porous sheets are originally obtained from





**Fig. 2.** The framework of the present deep learning model. (A) Two-step strategy utilized in the present deep learning framework. (B) The overall framework of the cGAN, which is powered by a Generator and a Discriminator. The chart is an illustrative example of predicting crack paths. The input to Generator 2 is a series of images of the SED field. Correspondingly, the output is a series of images containing crack information. (C) The architecture of the Generator. Both Generator 1 and Generator 2 use the same U-Net 3+ architectural framework. This framework consists of an encode-decode structure (colored blocks) and full-scale internal skip connections (arrows). Both the encoder and decoder contain nine layers, which are used to extract geometric feature maps at different scales. (D) The architecture of the Discriminator. The real and fake images come from the PFM and Generators, respectively.

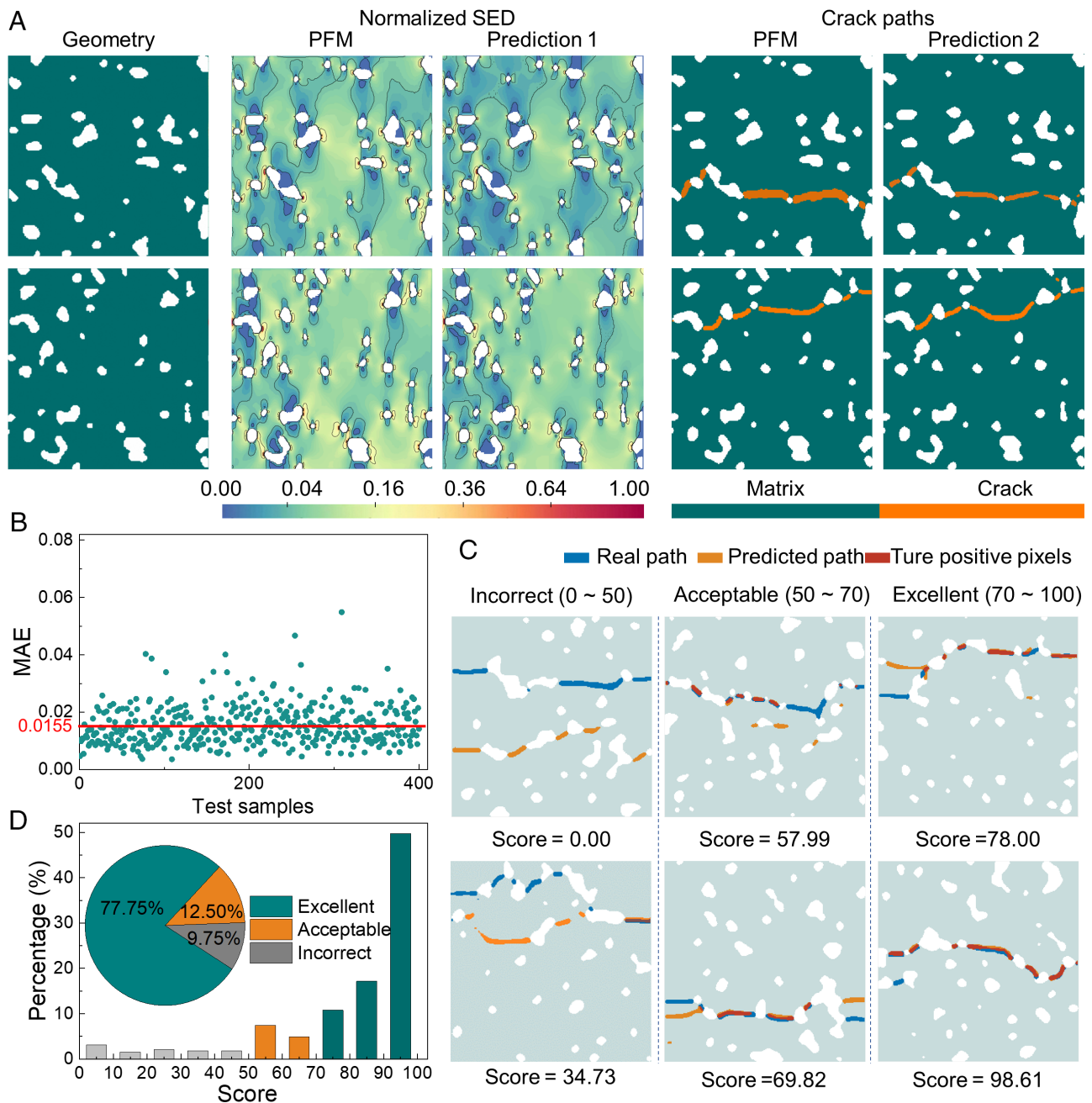
the photographic images. Subsequently, their crack paths are predicted via the present deep learning model. The procedure of deep learning prediction is shown in Fig. 4D and Movie S1, and the corresponding experimental preparations are detailed in the *Materials and Methods*.

Case 1: Three porous sheets of alumina ceramic ( $\text{Al}_2\text{O}_3$ ) with randomly distributed circular pores are prepared, as shown in Fig. 4E. The crack paths of these samples can be obtained simultaneously by the deep learning model and experiment (Movie S2). To ascertain the porous structures of the ceramic samples, photographs

of their effective regions are taken and the corresponding color schemes are modified. Specifically, in line with the conventions of the training dataset, the white and dark blue colors are used to denote the pores and the substrate, respectively. The predicted crack paths are in good consistency with the experimental observations. The present model accelerates the efficient of evaluating fracture response in porous structures and is beneficial to the design of high-performance crack-resistance materials.

Case 2: The core technology of extracting shale gas from porous rocks lies in fracturing the formation to establish the



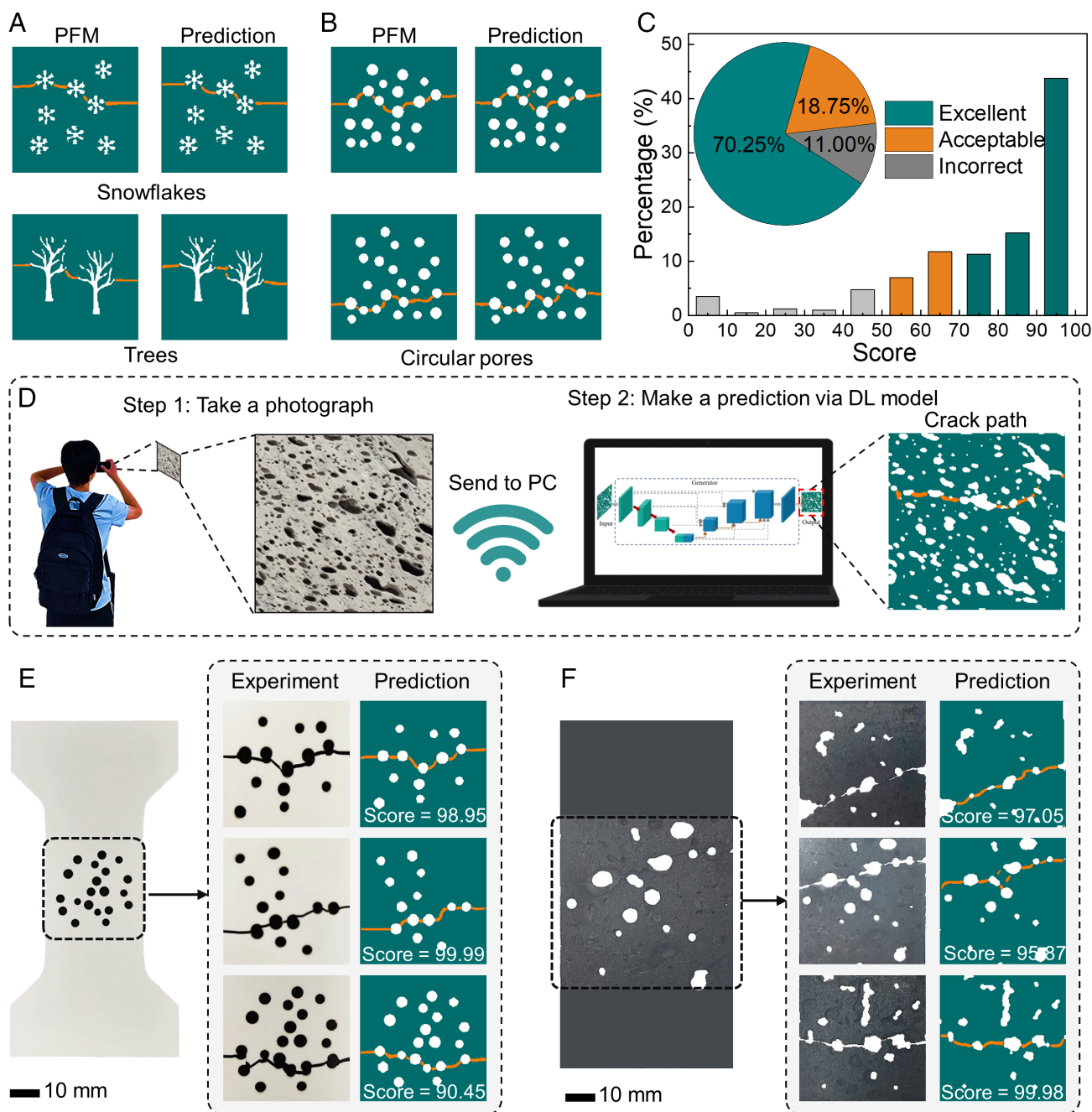


**Fig. 3.** Evaluation of the model performance. (A) Two representative predictions of normalized SED and crack paths. A nonlinear gradient “BlueRed” color map is used to visualize the distributions of SED around the pores. (B) MAEs for SED of all 400 test samples. Each point represents the MAE result corresponding to a sample in the test dataset. The average score for all test samples is denoted by the red line. (C) Representative predicted crack paths of three categories (“Incorrect,” “Acceptable,” and “Excellent”). To highlight the morphology of the cracks, the color of the matrix is represented by a light color (Red, Green and Blue (RGB) = [220, 230, 230]). The predicted crack paths and the simulation counterparts (PFM) are denoted by blue and orange, respectively. The overlapping pixels (true positive pixels) are marked by red. (D) Statistical results of the scores pertinent to the predicted crack paths for 400 samples in the test dataset. The pie chart represents the statistical results of the three categories.

interconnected microchannels for fluid transportation, such as hydraulic fracturing. To understand the failure of porous rocks, the present model is applied to predict their crack propagation. Three test samples of porous basalt with naturally occurring irregular pores are manufactured. Identical to Case 1, the crack paths of these samples are predicted by the deep learning model and observed by experiment. As exhibited in Fig. 4F, numerical predictions coincide with the experimental observations. The present study provides an avenue for quick evaluation of the fracture behaviors of porous rocks. The predicted results are beneficial to planning and controlling the crack propagation in porous formations.

## Discussion

In this work, a unique generative deep learning model is developed to accurately predict the crack paths in random porous media directly from their images. Owing to the fast inference capability of the neural network, tedious physical computations and experimental processes are skipped. The time of a fracture prediction is shortened from hours to a second in comparison with the traditional numerical model (*SI Appendix, Fig. S7*). This vastly speeds up the efficiency of evaluating fracture behaviors of heterogeneous materials with complicated geometries. The present model can also be extended to some customized structures with unique



**Fig. 4.** Generalization capability and application of the present deep learning model. (A) Predictions of crack paths in the customized patterns (Snowflakes and Trees). (B) Representative predictions of porous sheets with randomly distributed circular pores. The training data are composed of porous sheets with irregular random pores, while the test samples possess circular pores. (C) The statistical results of the scores pertinent to the predicted crack paths. The average score and accuracy (including Excellent and Acceptable) are 78.67 and 89.00%, respectively. The score is obtained based on the modified Hausdorff distance. (D) Flow chart of predicting crack path of the porous sheet directly from its photograph. (E) Ceramic sheets with randomly distributed circular pores. Crack paths are predicted by the present deep learning model and by experiment. (F) Experimental results and deep learning predictions of porous basalt sheets with natural porous structures.

geometric features. This merit is beneficial to the designs of high-performance crack-resistant porous media in conjunction with optimization algorithms.

As a retrospective study, we revisit the fracture problems of two porous samples with a high level of similarity of 99.54%. Owing to a small perturbation, the corresponding crack takes a quite different path. This subtle difference can be successfully captured by the present deep learning model (*SI Appendix, Fig. S2*). It indicates that the neural network grasps the intrinsic physical mechanism of fracture in porous media. Since fracture is one of the most challenging topics in solid mechanics, the present framework can be employed to predict other mechanical behaviors of porous

media at various stages of external loading, such as elastic deformation, plastic behavior, and damage.

To emphasize the necessity of intermediate prediction (SED) in the proposed deep learning framework, a single Generator model is trained to directly translate the microstructures into crack paths. With the simplified model, the prediction accuracy drops to 81.25% (*SI Appendix, Fig. S8*). This implies that directly establishing an implicit map between the geometry and final crack morphology is a difficult task. The introduction of the intermediate auxiliary variable SED strengthens the inference capability of the neural network. If Generator 2 takes the images of the precise SED field (calculated by the PFM) as input,

correspondingly the prediction accuracy of crack paths would further increase to 92.75%.

We also conduct a comprehensive analysis on the Incorrect predictions. For a limited number of test samples, the deep learning model may be deceived by spurious fracture patterns. From the perspective of the SED field, the fake crack path may possess an almost identical energy density to the real counterpart. To eliminate the interference of the fake crack paths, it is necessary to perform accurate mechanical calculations of the elastic responses and fracture parameters in porous media. The required high inference accuracy may far exceed the capabilities of image convolution operations. Consequently, it is difficult to improve the accuracy to the next level even with the most advanced deep learning techniques in computer vision. A purely data-driven approach may need to be modified accordingly, such as embedding the physical constraints relevant to the fracture problem. One promising solution is the physics-informed neural networks. Some attempts have been reported in recent works (43, 44) for some simple geometries.

## Materials and Methods

**Reconstruction of Random Porous Media.** The digital model of random porous media is composed of  $256 \times 256$  square pixels of size  $a^2$ , with  $a = 0.05$  mm being the length of pixels hereafter. The model can be described by a two-phase function  $Z(\mathbf{r})$ , with  $Z(\mathbf{r}) = 0$  and  $Z(\mathbf{r}) = 1$  representing the matrix and pores phase, respectively. The parameter  $\mathbf{r}$  denotes the position vector. The volume fraction and the characteristic size of the pores are represented by the porosity  $p$  and the correlation lengths  $l_c$ , respectively. In this work, the porosity  $p$  varies from 0.05 to 0.30, and the correlation lengths  $l_c$  are set to  $4a$ ,  $6a$ , and  $8a$ . The numerical algorithm of the reconstruction method may refer to ref. 39.

**Preparation of Dataset.** The cracks of reconstructed porous sheets are generated by tensile displacement loading. A PFM for fracture is used to simulate the quasistatic fracture response of reconstructed porous media. The derivation of this method is detailed in *SI Appendix, Text*. The PFM simulations are performed on the Chengdu Supercomputing Center Operations and Management Co. Ltd. The mesh convergence in the numerical calculations is discussed, and the present mesh level is justified (*SI Appendix, Fig. S9*).

Reconstructed geometric images and simulation results of the PFM, including elastic SED field and crack paths, are collected to compose the dataset. The map of elastic SED is first extracted at the elastic deformation stage (the strain  $\varepsilon = 0.1\%$ ) and then normalized by its maximum value. The dataset contains a total of 2,000 image pairs, 80% (1,600) of which serve as the training dataset and the remaining 20% (400) acts as the test dataset.

**Deep Learning Approach and Training Process.** Two independent cGANs with the same architecture are trained. The Generators (U-Net 3+) consists of an encoder, a decoder, and the internal full-scale skip connections, as illustrated in Fig. 2C. Each block of the encoder contains a convolutional layer with a kernel size of four and a stride of two. Batch normalization follows each convolutional layer, except for the first block. The nonlinear activation function of Leaky ReLU is then applied to each block. The Encoders transform a high-dimensional  $256 \times 256 \times 3$  pixel-image into a lower-dimensional  $1 \times 1 \times 512$  vector, which contains encrypted information and features of the input image. The Decoder is responsible for translating the encrypted information of the lower-dimensional feature maps to the target output image. Each block of Decoder receives feature maps of all scales through the full-scale skip connections (*SI Appendix, Fig. S10*). The fused multiscale feature maps are up-sampled by a transpose convolutional

layer. Batch normalization followed by the ReLU activation is then applied. A drop-out of 0.2 is applied for the first three blocks of Decoder to boost the robustness of the model.

The Discriminator (PatchGAN) consists of eight layers (Fig. 2D). The image inputting to the Discriminator is mapped to a  $30 \times 30$  matrix by the convolution operation. Each point in the matrix represents a patch in the original image. PatchGAN then determines the probability that each position in the matrix is true. The final output of the Discriminator is the average of the probabilities associated with all the patches.

The MAE (L1 loss) is used as the training objective function. The adaptive moment estimation (Adam) algorithm is adopted as the optimizer, and the learning rates are set to  $2 \times 10^{-4}$  and  $1 \times 10^{-5}$  for the Generators and Discriminators, respectively. The batch size is set to 16. The model is trained on a single NVIDIA Quadra A100 MIG and is terminated at the 1000<sup>th</sup> epoch. The whole process usually takes 33.5 h. The loss curves for both deep learning model of SED and crack paths as functions of iteration epochs are shown in *SI Appendix, Fig. S11*.

**Fabrication of Random Porous Samples.** An ultraviolet laser cutter is adopted to cut the customized porous structures in PMMA sheets of a thickness 1 mm (*SI Appendix, Fig. S12A*). The area of the porous specimen is  $20 \times 20$  mm<sup>2</sup>. The geometries of these specimens are generated by the method in Section Reconstruction of random porous media. The porosity  $p$  and correlation length  $l_c$  of the specimen are set to 0.1 and  $8a$ , respectively. The Young's modulus  $E$ , Poisson's ratio  $\nu$ , and critical energy release rate  $G_c$  are experimentally measured to be 2.3 GPa, 0.3 and 0.7 N/mm, respectively.

The perforated alumina ceramic samples are made from ceramic sheets of thickness 1 mm (*SI Appendix, Fig. S12B*). The effective area of the porous sample is  $20 \times 20$  mm<sup>2</sup>. The numerical algorithm for generating randomly distributed circular pores can be found in ref. 45. The diameters of the circular pores range from 1 mm to 3 mm.

The randomly distributed pores in porous basalt are naturally generated. In Fig. 4F, the characteristic lengths of pores vary from 2 mm to 5 mm. The rocks are separated to thin slices with a thickness of 4 mm to ensure the pores penetrate the samples, each of which may be approximated as a planar porous structure.

**Measurement of Fracture Responses.** The measurement is conducted on a micromechanical test system (*SI Appendix, Fig. S13*). Each thin porous sample is fixed with the rigid clamps at the top and bottom boundaries. A vertical upward displacement is applied on the top clamp at a velocity of 10  $\mu$ m/s. During the stretching process, the corresponding force-displacement curve is recorded by a mechanical sensor. The experiment is terminated when the sample completely fractured, and the corresponding crack path is recorded via a photograph.

**Data, Materials, and Software Availability.** Code and data (image dataset associated with random porous media) have been deposited in Open Science Framework (DOI: [10.17605/OSF.IO/WVZYD](https://doi.org/10.17605/OSF.IO/WVZYD)) (46).

**ACKNOWLEDGMENTS.** We express our gratitude to Dr. Jielei Chu for his insightful suggestions regarding the deep learning model. The constructive discussions with Chang Liu, Jiao Wang, Xuebo Yuan, Yayu Li and Bing Li from Southwest Jiaotong University are appreciated. This work was supported by National Natural Science Foundation of China (Nos.: 12072297, 12192210, 12192211, and 12192214) and the support from the New Interdisciplinary Cultivation Funds of Southwest Jiaotong University (No.: 2682024KJ013) is acknowledged as well.

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